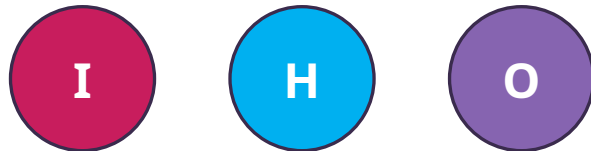


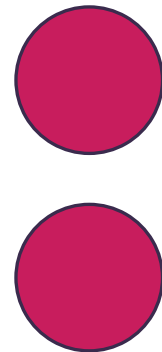
Neural Network basics



- Computer neural networks may take on some characteristics of a cluster of neurons, but the similarity is only superficial
- Neural networks are made up of multiple layers of interconnected nodes that can learn to perform tasks by analyzing examples, without being explicitly programmed
- Each node has a numerical value assigned to it and is thought of as a neuron
- Networks are composed of multiple node layers: Input, Hidden, Output

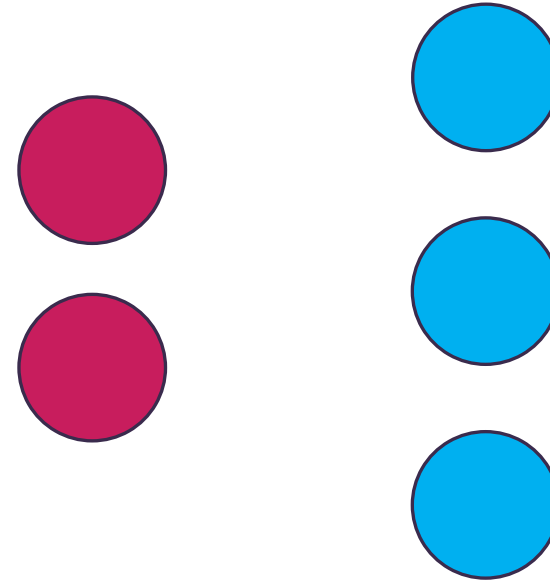
Neural Network construction

- Input Layer nodes receive numerical data (typically) for processing by the Network; these nodes values are supplied in order to recall a previously remembered value



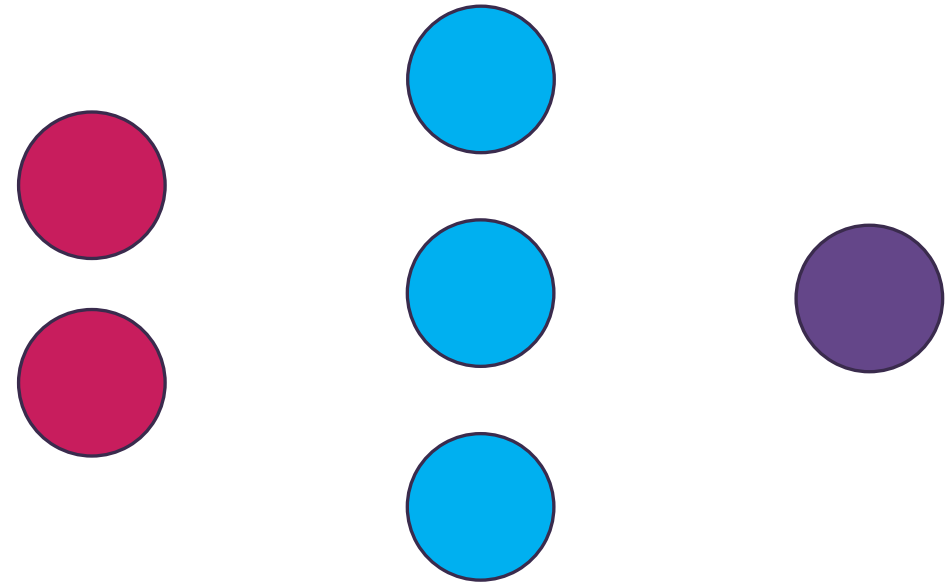
Neural Network construction, cont'd

- Input Layer nodes receive data for processing by the Network
- Hidden Layer nodes are intermediate nodes between the Input and Output layers; these nodes form the engine of the network and are instrumental to the performance of the network



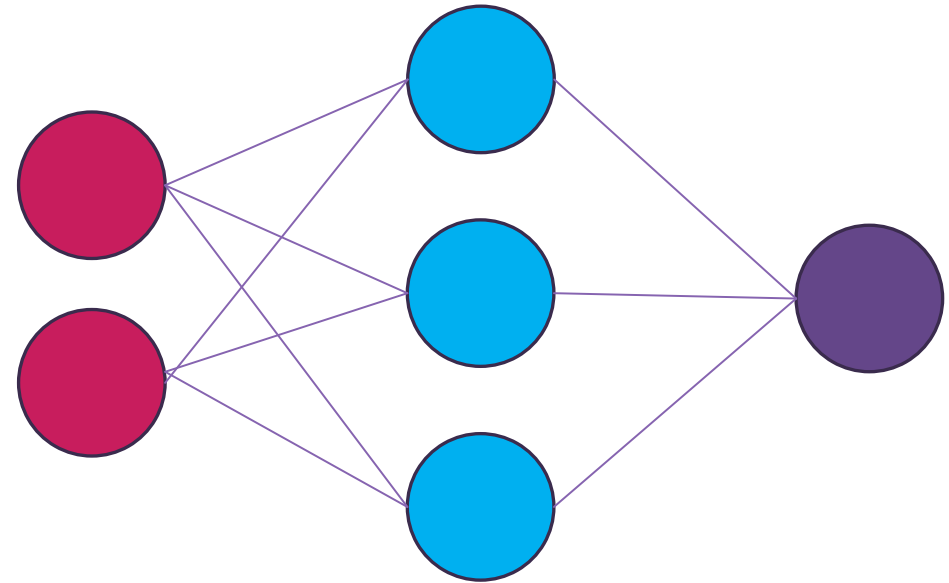
Neural Network construction, cont'd

- Input Layer nodes receive data for processing by the Network
- Hidden Layer nodes are between the Input and Output layers
- Output Layer nodes contain the resulting values after the input nodes are processed; these contain the “recalled” memory from the Network



Neural Network construction, cont'd

- Input Layer nodes receive data for processing by the Network
- Hidden Layer nodes are between the Input and Output layers
- Output Layer nodes contain the resulting values
- Numerical weights connect the Nodes where each Node in a Layer is connected to every other Node in the next subsequent Layer



Neural Network construction, cont'd

- Input Layer nodes receive data for processing by the Network
- Hidden Layer nodes are between the Input and Output layers
- Output Layer nodes contain the resulting values
- Weights connect the Nodes
- Weights are modified during Training, as with the Node values, to strengthen pathways for the successful recall of information

